# AUTOENCODER-BASED IMAGE COMPRESSION


## ABSTRACT

Image compression is an important technique used to reduce the storage space required for digital images. Traditional compression methods use predefined algorithms, whereas deep learning can learn useful image representations from training data. This project proposes an Autoencoder-Based Image Compression system using deep neural networks. An autoencoder consists of an encoder, a latent representation, and a decoder. The encoder converts an image into a compact representation, while the decoder reconstructs the image from that representation. The reconstructed image is compared with the original image using evaluation metrics such as Mean Squared Error (MSE) and Peak Signal-to-Noise Ratio (PSNR). This project demonstrates the application of deep unsupervised learning to image representation and reconstruction.



**Keywords:** Deep Learning, Autoencoder, Image Compression, Unsupervised Learning, Neural Networks.


## 1. INTRODUCTION

Digital images are widely used in mobile applications, websites, medical systems, and communication technologies. High-quality images require more storage space and bandwidth. Image compression helps reduce the amount of data needed to represent an image.

Deep learning provides an alternative approach by learning image features automatically. An autoencoder is a neural network that learns to reconstruct its input. It can learn a compact representation of an image and use that representation to generate a reconstructed output.

This project explores how autoencoders can be applied to image compression and reconstruction.

## 2. PROBLEM STATEMENT

Large digital images require considerable storage space and may take longer to transfer over networks. Conventional compression techniques may not preserve every image feature equally.

The objective of this project is to study whether a deep-learning-based autoencoder can learn compact image representations and reconstruct images with acceptable quality.

## 3. OBJECTIVES

1. To understand the concept of autoencoders in deep learning.
2. To design an encoder and decoder neural network.
3. To prepare and preprocess an image dataset.
4. To train an autoencoder to reconstruct images.
5. To evaluate reconstruction quality using MSE and PSNR.
6. To study the relationship between representation size and image quality.

## 4. SCOPE OF THE PROJECT

This project focuses on image reconstruction using a convolutional autoencoder. A publicly available image dataset, such as CIFAR-10, can be used for experimentation.

The project demonstrates the basic principles of learned image representations. It is an educational prototype and does not claim to replace established image-compression formats.

**5. INTRODUCTION TO AUTOENCODERS**

An autoencoder is a neural network trained to reproduce its input at its output. It learns useful features by minimizing the difference between the original input and reconstructed output.

An autoencoder contains three main components:

**Encoder:** Converts the input image into a smaller or more compact representation.

**Latent Representation:** Stores the important features learned from the input image.

**Decoder:** Uses the latent representation to reconstruct the original image.

Autoencoders can learn image patterns without requiring a separate class label for every image.

**6. PROPOSED SYSTEM**

The proposed system uses a convolutional autoencoder for image reconstruction.

The workflow consists of the following stages:

1. Collect a suitable image dataset.
2. Resize images to a consistent dimension.
3. Normalize pixel values.
4. Divide the dataset into training, validation, and test sets.
5. Train the autoencoder using the training images.
6. Generate reconstructed images.
7. Compare reconstructed images with the original images.
8. Evaluate the model using suitable metrics.

**System Workflow:**

Input Image → Preprocessing → Encoder → Latent Representation → Decoder → Reconstructed Image → Evaluation

**7. DATASET PREPARATION**

A publicly available image dataset such as CIFAR-10 can be used. The selected images are resized to a consistent input dimension and normalized before training.

The dataset is divided into training, validation, and test sets. The training set is used to learn model parameters, the validation set is used to monitor training, and the test set is reserved for final evaluation.

The dataset source, number of images, image dimensions, and split proportions should be recorded during implementation.

**8. AUTOENCODER ARCHITECTURE**

The proposed architecture consists of the following layers:

**Input Layer:** Receives the original image.

**Convolutional Layers:** Extract visual features such as edges, textures, and colour patterns.

**Downsampling Layers:** Reduce the spatial dimensions of the feature maps.

**Latent Layer:** Represents the extracted information in a compact form.

**Decoder Layers:** Reconstruct the image from the latent representation.

**Output Layer:** Produces the reconstructed image with the same dimensions as the input image.

The model learns by comparing the reconstructed output with the original image and updating its weights through backpropagation.

**9. TRAINING METHODOLOGY**

The autoencoder is trained using a reconstruction loss function. Mean Squared Error is a suitable choice because it measures the average squared difference between corresponding pixel values in the original and reconstructed images.

An optimizer such as Adam can be used to update the network parameters. During training, validation loss is monitored to identify overfitting and determine when training should stop.

Important hyperparameters include learning rate, batch size, number of epochs, latent dimension, optimizer, and loss function.

**10. EVALUATION METRICS**

The performance of the autoencoder can be evaluated using the following measures.

**Mean Squared Error (MSE):** Measures the average squared difference between original and reconstructed pixels. Lower values generally indicate more accurate pixel-level reconstruction.

**Peak Signal-to-Noise Ratio (PSNR):** Measures reconstruction quality in decibels. Higher values generally indicate lower reconstruction error for the same image data range.

**Visual Comparison:** Original and reconstructed images are compared to identify loss of details, blurring, and colour changes.

**Latent Representation Size:** The dimensions of the latent representation are recorded to study the compactness of the learned representation. This alone is not the same as the actual compressed file size.

**11. RESULTS AND DISCUSSION**

The model should be evaluated after training using the reserved test dataset. The original and reconstructed images can be displayed side by side to observe the quality of reconstruction.

The following results should be recorded after the experiment:

=>Dataset used and number of test images.

=>Final test MSE.

=>Average test PSNR.

=>Latent representation dimensions.

=>Visual differences between original and reconstructed images.

Actual numerical results must be obtained by running the model. They should not be estimated or invented.

**12. APPLICATIONS**

Autoencoder-based image reconstruction has potential applications in:

1. Image representation learning.
2. Image storage research.
3. Image preprocessing for machine-learning applications.
4. Educational demonstrations of deep neural networks.
5. Research into learned image-compression techniques.

Practical compression applications require additional testing and comparison with established compression methods.

**13. ADVANTAGES**

=>Learns useful image features automatically.

=>Does not require class labels for reconstruction training.

=>Can learn compact representations of image data.

=>Supports experimentation with different latent dimensions.

=>Demonstrates the use of convolutional neural networks in unsupervised learning.

**14. LIMITATIONS**

=>Reconstruction quality depends on the dataset and network architecture.

=>Important details may be lost when the latent representation is too small.

=>Training requires computational resources.

=>MSE and PSNR do not completely represent human perception of image quality.

=>A compact latent representation does not automatically produce a smaller encoded image file.

=>The model may not perform equally well on unfamiliar image types.

**15. FUTURE ENHANCEMENTS**

The project can be extended by comparing different latent dimensions and autoencoder architectures. A denoising autoencoder can be explored to reconstruct images corrupted by noise. Future work can also include perceptual loss functions, a simple image-upload interface, and comparisons with traditional image-compression methods using actual encoded file sizes.

**16. CONCLUSION**

This project studies autoencoder-based image compression using deep unsupervised learning. The encoder learns a compact representation of an image, and the decoder reconstructs the image from that representation. Reconstruction quality can be evaluated using MSE, PSNR, and visual comparison. The project provides an understanding of autoencoder architecture, training, and image representation learning. Final conclusions about performance should be based on measured experimental results.

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